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With or Without Forecast Sharing: Competition and Credibility under Information Asymmetry

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Forecast sharing among trading partners lies at the heart of many collaborative and contractual supply chain management efforts. Even though it has been praised in both academic and practitioner circles for its critical role in increasing demand visibility, some concerns remain: The first one is related to the credibility of forecast sharing, and the second is the fear that it may turn into a competitive disadvantage and induce suppliers to increase their price offerings. In this study, we explore the validity of these concerns under a supply chain with a competitive upstream structure, focusing specifically on (i) when and how a credible forecast sharing can be sustainable, and (ii) how it impacts on the intensity of price competition. To address these issues, we develop a supply chain model with a buyer facing a demand risk and two heterogeneous suppliers *competing* for order allocation from the buyer. The extent of demand is known only to the buyer. The buyer submits a buying request to the suppliers via a commonly used procurement mechanism called request for quotation (RFQ). We consider two variants of RFQ. In the first type, the buyer simply shares the estimated order quantity with no further specifications. In the second one, in addition to this, the buyer also specifies minimum and/or maximum order quantities. We fully characterize equilibrium decisions and profits associated with them under symmetric and asymmetric information scenarios. Our main findings are that the buyer can use a RFQ with quantity restrictions as a credible signal for forecast sharing as long as the degree of demand information asymmetry is not too high, and that, *contrary* to above concerns, the equilibrium prices that emerge between competing suppliers under asymmetric information may indeed *increase* if the buyer can*not* share forecast information credibly with its upstream partners.

Key words: forecast sharing; competition; credibility; asymmetric information; signalling
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1. Introduction

Suppliers and buyers often have to make pricing and sourcing decisions, respectively, without knowing how much supply and demand will actually materialize. This inevitably leads to an excess in supply or demand. In its quest to reduce the detrimental effects of supply–demand mismatches on overall chain performance, supply chain management (SCM) is continually pushing boundaries by considering numerous procurement mechanisms with varying degrees of integration and sophistication. Among the collaborative tools considered are Electronic Data Interchange, Efficient Consumer Response, and Quick Response, all of which serve a common objective: to enhance *demand visibility* across the chain. In most B2B transactions, the buyer initiates the procurement process by issuing a buying request to a set of pre-screened suppliers. We observe various forms of such requests used in practice such as request for proposal, request for bid, and request for quotation (RFQ). In this study, we focus on the latter and shortly refer to it as "RFQ."

A standard RFQ typically specifies the information about the items/services to be procured, estimated order quantity, and expiration time.[1] In addition to this, the buyers may include additional specifications such as minimum and maximum order quantities to provide more information to the suppliers about his demand. For example, a sample RFQ generated based on UN/EDIFACT standards[2] may stipulate lower and upper bound constraints on the amount of products to be procured from the supplier who wins the bid. Upon receiving the RFQ from the buyer, the suppliers have to respond before the expiration date by submitting their quotations which typically include the price per item/service. After assessing the suppliers' quotations based on their costs and risks, the buyer selects the winner(s), with whom he engages in a procurement contract. At this stage, there are various forms of contractual mechanisms that the buyer can use to implement minimum and/or maximum order specifications such as quantity flexibility and take or pay. Agreed between buyers and suppliers, these contracts help the buyer to specify the minimum ("take") and maximum ("pay") quantities that a buyer can order from the supplier (see Bassok et al. 1997, Farlow et al. 1996b, Masten and Crocker 1985 for the practical uses of these contracts in industries such as the automobile industry, semiconductors, electronic equipments, energy, and natural resources).

In SCM, the potential benefits of sharing forecast information to the chain are rarely questioned. Those benefits are mainly attributable to by-products of increased demand visibility (Helms et al. 2000, Seifert 2003). Among the most frequently cited benefits are a reduction in inventory-related costs, an increased response time, and improved service levels (Schachtman 2000, Småros 2007). While the benefits of forecast sharing are impressive, its actual implementation offers challenges that hinder its adoption (Barratt 2004). In a recent survey of 120 companies, conducted by Fraser (2003), 42% of all respondents listed a lack of trust in sharing sensitive demand information as one of the top three challenges they faced with this matter.[3] Concerns about the lack of trust can arise for many reasons, including, high demand uncertainty, risk due to potential loss from trusting (Özer et al. 2011), a potential loss of control, or security breaches (Fliedner 2003). In this study, we will focus on two reasons.

The first reason we will study is the credibility of forecast sharing in a competitive supply chain environment. As demonstrated in various studies, a forecast information asymmetry among supply chain parties naturally creates an incentive for information distortion to influence the decision of uninformed parties. This in turn raises question marks in the un informed parties' minds regarding the credibility of the forecast information provided to them. Indeed, prior to adopting specific mechanisms that involve further restrictions on the order quantities, Sun MicroSystems was simply sharing its estimated demand information with its suppliers with no commitment on either party. This in turn made a "dissatisfaction with Sun's component forecasts," the number one issue in the voice-of-the-supplier feedback collected from the suppliers (Farlow et al. 1996a). Planning decisions made with a distorted picture of the actual demand information may create significant inefficiencies at every point along the supply chain, which can potentially hurt the end consumers, causing poorer customer service and higher unit costs (Lee et al. 1997).

Our second concern is just the opposite of the above one. Indeed, it is the fear that *truthful* forecast sharing may adversely affect the degree of competition in a supply chain at the expense of the party who shares the forecasts with the supply chain partners. Stein (1998) describes this fear as follows:

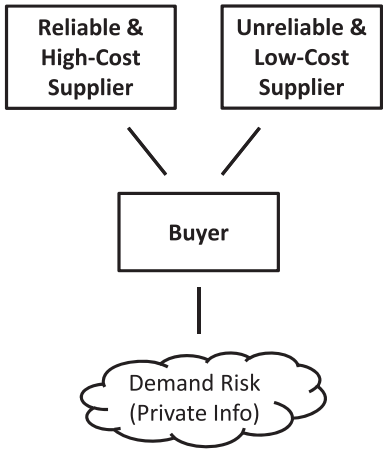
> Few organizations are accustomed to trusting their business partners, and they have a real and justified fear that information sharing can turn into a competitive disadvantage. "If one of your customers knows you're behind on your production schedule, they may turn around and try to negotiate a more favourable price," says Harry Tse, an analyst with the Yankee Group Inc.

These concerns are also echoed by Verity (1996) and Fliedner (2003). They note that *by coordinating forecasts, two or more suppliers can influence the supply and drive up its price*. Fliedner (2003) adds that such an event is more likely to happen when the item being purchased is custom-made or of a proprietary nature, making it less readily available.

Even though forecast sharing has been extensively explored in the literature, to the best of our knowledge, the credibility of forecast sharing and its impact on the degree of upstream competition have not received due attention for competitive supply chains. Along these lines, the objectives of this study are threefold: (i) to explore *when* and *how* forecast information can be credibly shared across a supply chain with a *competitive upstream* structure, (ii) to identify how this impacts the degree of upstream competition, and finally (iii) to evaluate the impact of forecast sharing on the decisions/profits/costs of channel partners.

To address these issues, we develop a *bi-level* supply chain model that captures the salient features of industries that would potentially benefit from forecast sharing (i.e., high supply and demand mismatch risk, and asymmetric demand information). See Figure 1 for the high-level representation of the supply chain model considered in this study. At the upstream level, there are *two competing* heterogeneous sellers (hereinafter referred to as "supplier U" and "supplier R") with different costs and risk structures. At the downstream level, there is one buyer who faces a risky demand, the extent of which is known only to the buyer himself.[4] First, the buyer issues a buying request in the form of a RFQ. We analyze two variants: non-restrictive and restrictive that will be referred to as RFQ^n and RFQ^r, respectively. In the former, the buyer only specifies the estimated demand

Figure 1 Supply Chain Model Considered in this Study



information, whereas in the latter, in addition to the estimated demand information, the buyer also specifies a lower and/or upper bound constraint on the amount of products to be ordered. In response to the RFQ, the suppliers compete on prices, and submit their quotations (i.e., "bids"). After assessing the suppliers' bids, the buyer makes his final order allocation decision. Finally, in the last stage, the supply and demand uncertainties are realized and the buyer clears the realized supply–demand mismatch, if needed.

We characterize equilibrium RFQ terms under each RFQ type as well as prices associated with them under both the symmetric and asymmetric information scenarios. Then, the comparative analysis of RFQ^n and RFQ^r under symmetric and asymmetric information settings allows us to evaluate when and how the forecast information can be credibly shared between channel partners and how it affects the degree of upstream competition and the profits and costs.

First of all, the suppliers' incomplete information about the demand leads to an uncertainty on their side regarding what price to charge, which, in turn, generates an incentive for the buyer to distort his forecast information, and invalidates the credibility of the estimated demand information in RFQ^n. We show that this distortion can be fixed via an RFQ^r as long as the degree of information asymmetry is not too high. Second, in the symmetric information setting, when one of the suppliers undercuts her opponent's price, she does so by taking into account the true demand scenario of the buyer. However, in the presence of incomplete demand information, the suppliers cannot credibly infer the true demand scenario, which enforces them to decide on their prices solely based on their *a priori* beliefs. In this case, each supplier makes two choices: (i) whether to undercut her opponent for all possible demand scenarios (*mass-market strategy*) or for a particular one (*niche-market strategy*), and (ii) whether to compete for the same demand scenario(s) (*head-to-head price competition*) or split the demand scenarios between themselves. Depending on the choices made by each supplier, the strength of price competition (resp. equilibrium price) decreases (resp. increases) from one extreme to another as the suppliers move from mass- to niche-market, and from head-to-head price competition for the same market to splitting the market among themselves. Our results show that RFQ^n pushes the degree of price competition to the extremes at either end, whereas RFQ^r keeps it in the middle. This result contradicts the concerns echoed by supply chain practitioners regarding the potential role of forecast sharing in driving prices up. In other words, credible forecast sharing can indeed help the buyer to pull the strings of suppliers and align them to compete for the same forecast, which results in lower prices. Finally, a comparison of costs and profit under both types of RFQs enables us to evaluate when a restrictive RFQ type is a better option [and when it is not] for each supply chain partner.

This article is organized as follows: in the next section, we review the relevant literature. In section 3, we develop the model framework. In sections 4 and 5, we analyze the symmetric and asymmetric information scenarios. In section 6, we compare the impact of each RFQ type on the equilibrium decisions, profits, and cost of channel partners. Section 7 concludes.

2. Relevant Literature

There are two streams of research in the OM literature that consider the impact of sharing forecast information on supply chains.

Articles in the first stream assume that supply chain partners make their decisions in a cooperative manner (i.e., that they all minimize the same system-wide costs), but that they differ in terms of the amount of demand information that they have before they make operational decisions. In general, asymmetric demand information between supply chain partners leads to a variance increase in the demand in the upstream direction, a phenomenon known as the *bull-whip effect* (Lee et al. 1997). Several models have been developed to quantify the magnitude of the bull-whip effect under various demand models: Integrated Moving Average (IMA) demand model in Graves (1999), Auto-Regressive (AR) demand model in Lee et al. (2000), and Auto-Regressive Moving Average (ARMA) in Gaur et al. (2005). Chen et al. (2000) explore the relation between two commonly used forecasting techniques (moving average and exponential smoothing) and the bull-whip effect. Gavirneni et al. (1999) consider a capacitated supply chain setting and explore the impact of sharing forecast information on inventory policies. Another reason for the bull-whip effect is the lack of coordination between supply chain firms regarding forecast generation. Aviv (2001) considers a supply chain where local and collaborative forecasting techniques are used to decide inventory orders, then compares these types of techniques in a setting with stationary and auto-correlated demand distributions.

The second stream of research explores the value of forecast sharing in decentralized supply chains. We refer the readers to Chen (2003) for a comprehensive review of the impact of information sharing on decentralized supply chains. Among the articles in this stream, Cachon and Lariviere (2001) are the first to analyze the buyer's incentives to share demand

forecast information and the role of compliance regimes in a dyadic supply chain setting. Li (2002) and Zhang (2002) consider a supply chain structure that consists of one manufacturer selling to two retailers. They analyze the impact of forecast sharing on the retail (downstream) competition, which contrasts with our study. Li and Zhang (2008) extend this to a supply chain setting where a single manufacturer sells to multiple retailers. Ha and Tong (2008) and Ha et al. (2011) consider a similar setting with two competing supply chains, each of which has one retailer and one manufacturer. In all these articles (with the exceptions of Cachon and Lariviere 2001, Özer and Wei 2006, Oh and Özer 2013), it is assumed that if the parties decide to share their private forecast information, they do so *truthfully*. Credibility, however, can be a legitimate concern if the parties have an incentive to distort their information. Cachon and Lariviere (2001) and Özer and Wei (2006) are the first to analyze the credibility of the forecast sharing in a dyadic supply chain setting with one manufacturer selling to one retailer. Using one manufacturer–one supplier setting, Oh and Özer (2013) analyze the role of time in forecast information sharing in a dynamic setting where firms' forecast information and hence, the asymmetry among such information evolves overtime. In addition to the analytical models, there are also two other growing streams in operations management that are related to our work: (i) empirical analysis of forecast sharing in supply chains (e.g., Bray and Mendelson 2012, Terwiesch et al. 2005) and (ii) behavioural research related to the role of trust in forecast sharing (e.g., Ebrahim-Khanjari et al. 2012, Özer et al. 2011). However, these streams differ from our article in terms of both methodology and setting.

Our study is also related to another stream of literature, which evaluates the impact of supply contracts with minimum and/or maximum order specifications on operational issues, such as information sharing (Cachon and Lariviere 2001, Lee et al. 1997, Özer and Wei 2006), variance of order-up-to-levels (Bassok and Anupindi 1997), exchange risk (Scheller-Wolf and Tayur 1997), coordination of supply chains (Tsay 1999), and inventory characteristics and fill rate (Bassok and Anupindi 2008). There is an abundant literature in terms of models that consider variants of contracts with constraints on the order quantity such as rolling horizon flexibility (RHZ) contracts (Bassok and Anupindi 2008), contracts that impose constraints on the total order quantity for single and multiple products (Anupindi and Bassok 1998, Bassok and Anupindi 1997, Chen and Krass 2001, Özer and Wei 2006), and take-or-pay agreements commonly used in the natural gas industry (Masten and Crocker 1985). The common feature among all these contracts is that the buyer guarantees ahead of time that his total order quantity will satisfy certain pre-determined thresholds. We refer the readers to Anupindi and Bassok (1999) and Tsay et al. (1999) for a review of supply contracts with quantity commitments.

The main objective of this study is to analyze both the credibility of forecast information in a competitive supply chain setting and the impact of forecast sharing on the degree of upstream competition. These features distinguish our model from the ones above. To address these issues, we explicitly consider a decentralized supply chain setting where (i) the buyer sources from two competing suppliers, (ii) the buyer faces private demand uncertainty, and (iii) the credibility of the demand information shared by the buyer is not assumed exogenously. This contrasts with the above articles, which consider a supply chain configuration where each buyer is served only by one supplier, and therefore analyze the credibility and/or the impact of forecast sharing on either downstream or vertical (i.e., supplier–buyer) competition.

Finally, we use a mixed modelling approach to model the impact of credible information sharing on the strength of upstream competition. Namely, we use a signalling game framework to model the RFQ-stage game between the buyer and the two suppliers, and a common agency framework to model the pricing-stage game between two suppliers (competing principals) and the buyer (common agent) (see Riley 2001, Martimort 2006 for extensive literature review on signalling games, and common agency problems, respectively). This distinguishes our modelling framework from the informed principal literature (e.g., Maskin and Tirole 1990, Mylovanov and Tröger 2012, Severinov 2008, Tan 1996) in three ways. First, in the models analyzed in this literature ("informed principal models with private values"), the informed party would never be hurt by its private information, whereas in our framework, as typical in a standard signalling game à la Spence (1973), it is not the case. In other words, the informed party (i.e., the buyer in our framework) needs to undertake costly signalling action to credibly convey his private information to the uninformed parties (i.e., the suppliers). Second, the revelation principle (Myerson 1979) can be employed to simplify the characterization of the optimal complete contract in informed principal models. However, it has long been known to be extremely challenging to characterize similar complete contracts for signalling games.[5] Finally, while the multiplicity of equilibria can generally be avoided in the informed principal literature, it is well known that both signalling and common agency games suffer from multiple equilibria due to the extra degrees of freedom caused by unrestricted off-equilibrium beliefs. For this reason, the main theme of these literatures has been

on the equilibrium selection (refinement) problem (please refer to Fudenberg and Tirole 1991, Martimort 2006).

3. Model Framework

To model the impact of credible forecast sharing on the degree of upstream competition, we consider a *bi-level decentralized* supply chain model with *asymmetric forecast information*. The supply chain consists of two competing suppliers at the upstream level and a buyer at the downstream level. The buyer is exposed to demand uncertainty, whose distribution is known only to him. On the other hand, the suppliers have prior beliefs on the true demand distribution. For the sake of analytical tractability, we assume that the demand distribution can be of two types; either high ($\phi = h$), or low ($\phi = l$). Let r^{ϕ} be the suppliers' prior beliefs for a ϕ-type demand distribution, where r^h and r^l are common knowledge and sum up to 1, i.e., $r^h + r^l = 1$. Let a ϕ-type buyer denote the buyer whose demand uncertainty is of type ϕ and ε^{ϕ} represent the demand uncertainty he faces. To develop analytical managerial insights, we assume that the ϕ-type buyer's demand uncertainty (shortly, the ϕ-type demand) has a two-point distribution, where low and high demand states are represented by 0 and Q, respectively.[6] Therefore, the demand uncertainty faced by ϕ-type buyer, i.e., ε^{ϕ}, can be expressed as follows:

$$\epsilon^{\phi} = \begin{cases} 0 & \text{with prob. } 1-\beta^{\phi} \\ Q & \text{with prob. } \beta^{\phi}. \end{cases} \tag{1}$$

where $\beta^{\phi} \in [0,1]$, $\phi \in \{l,h\}$, is the likelihood of the high demand state for ϕ-type demand. To make sure that an h-type demand is *larger* than an l-type demand in the sense of first-order stochastic dominance,[7] we also assume that $\beta^l \le \beta^h$.

Suppliers at the upstream layer engage in price competition and seek an order allocation from the buyer. However, they differ in terms of their exposure to yield risk and their cost structure.[8] The *unreliable* supplier (hereafter referred to as "U") has access to a production technology with random yield at a lower marginal cost of c_U per unit. So, if the buyer orders q_U units from supplier U, she delivers only $\tilde{q}_U$ units to the buyer, where, similar to the demand uncertainty, we assume "all-or-nothing"-type distribution for $\tilde{q}_U$, i.e.,

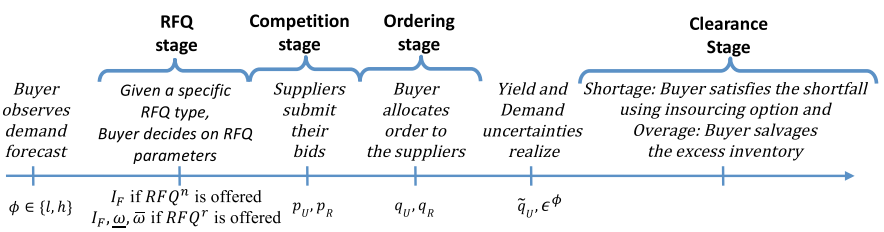
$$\tilde{q}_U = \begin{cases} 0 & \text{with prob. } 1-\alpha \\ q_U & \text{with prob. } \alpha \end{cases} \tag{2}$$

where α denotes the expected yield for supplier U. In contrast, the *reliable* supplier (hereinafter referred to as "R") has access to a fully reliable production technology (i.e., $\tilde{q}_R = q_R$ with prob. 1) albeit at a higher marginal cost of c_R per unit, where $c_U \le c_R$. That is, if the buyer orders q_R units from supplier R, she incurs $c_R q_R$ in total and delivers all the units to the buyer. Once demand and yield uncertainties are realized, the buyer pays to the suppliers only for the units delivered (in total, $p_R q_R$ and $p_U \tilde{q}_U$ for suppliers R and U, respectively), and clears the mismatch between realized demand and supply as follows. If the realized demand exceeds the total number of units delivered by the two suppliers, i.e., $\epsilon^{\phi} \ge \tilde{q}_U + q_R$, the buyer satisfies the shortfall from an insourcing option at a marginal cost c_I. The insourcing option[9] is assumed to have a zero lead time, be fully reliable, and be more costly than both suppliers. On the other hand, if the total realized supply exceeds total demand, the buyer salvages excess inventory $(\tilde{q}_U + q_R - \epsilon^{\phi})^+$ at c_S per unit, where we assume $c_S \le c_U$ to eliminate trivial cases. To simplify the analysis and presentation of our results, throughout this study, we further normalize marginal insourcing cost c_I to 1, marginal salvage value c_S to 0, and assume that $0 \le c_U \le c_R \le \beta^{\phi}$ for both $\phi \in \{l,h\}$.[10]

On the basis of the above framework, we then consider two RFQ types that differ only in terms of whether or not the buyer imposes lower and/or upper bound constraints on his initial estimated order quantity. Under each RFQ type, the three entities introduced above (i.e., the buyer, and two suppliers) interact with each other in the following *four stages*. The timing of decisions and events is illustrated in Figure 2.

1. The buyer observes his true demand type ϕ, where $\phi \in \{l,h\}$.

Figure 2 Timeline of Decisions and Events



2. *RFQ stage:* Depending on the type of RFQ, the buyer decides on the RFQ parameters. That is, if he offers an RFQ^n, he decides on I_F, whereas if he offers an RFQ^r, then he decides on $(I_F, \underline{\omega}, \bar{\omega})$, where I_F represents initial forecast and $\bar{\omega}$ and $\underline{\omega}$ correspond to, respectively, the upper and lower bounds on the total order quantity in terms of I_F.
3. *Competition stage:* In response to the buyer's RFQ parameters, suppliers U and R update their prior beliefs and *simultaneously* submit their unit prices (i.e., "bids"), p_U and p_R, respectively.
4. *Ordering Stage:* Based on the unit prices from the suppliers, the buyer then decides on his final order allocation decision. If an RFQ^r is offered, then the buyer's total order quantity should be between $I_F(1-\underline{\omega})$ and $I_F(1+\bar{\omega})$, i.e., $I_F(1-\underline{\omega}) \le q_U + q_R \le I_F(1+\bar{\omega})$.
5. Demand and yield uncertainties are realized, suppliers R and U deliver q_R and $\tilde{q}_U$ and receive in total $p_R q_R$ and $p_U \tilde{q}_U$ from the buyer, respectively.
6. *Clearance Stage:* If the total amount delivered by the suppliers turns out to be less than the realized demand, i.e., $q_R + \tilde{q}_U \le \epsilon^\phi$, then the buyer satisfies the shortfall, $[\epsilon^\phi - (q_R + \tilde{q}_U)]^+$, from the insourcing option.

Before analyzing the game, we summarize the list of notations used for the problem parameters and decision variables in Table 1.

Table 1 Notation used for Model Parameters and Decision Variables

Model parameters	
$\phi \in \{l,h\}$	Type of true demand distribution
r^ϕ; $\hat{r}^\phi$	Suppliers' prior and updated (posterior) beliefs about true demand type.
ε^ϕ; β^ϕ	Random variable denoting ϕ-type demand; the likelihood of high demand state for ϕ-type demand
$\tilde{q}_U$	Random variable denoting supplier U's output
α; $\bar{\alpha}$	Probability of no-disruption and disruption for supplier U, respectively, i.e., $\bar{\alpha} = 1-\alpha$
c_U;c_R	Marginal costs of suppliers U and R, respectively
Profit and cost functions	
π_U; π_R	Expected profits for suppliers U and R, respectively.
TC_B^ϕ	Expected cost for the buyer facing ϕ-type demand
Decision variables	
I_F^ϕ	RFQ^n parameter offered by ϕ-type buyer
I_F^ϕ; $\underline{\omega}^\phi$; $\bar{\omega}^\phi$	RFQ^r parameters offered by ϕ-type buyer.
p_U;p_R	Unit prices ("bids") quoted by suppliers U and R, respectively.
q_U^ϕ; q_R^ϕ	ϕ-type buyer's order allocations for suppliers U and R, respectively.

Throughout this study, equilibrium profits/cost and decision variables are annotated with asterisks ("*").

4. Equilibrium Analysis under a Symmetric Information Scenario

To establish a benchmark, in this section, we restrict our attention to the case where the true demand type is known by all the parties in the supply chain. This assumption transforms the problem to a symmetric information *Stackelberg* game between two competing suppliers at the upstream level and the buyer at the downstream level. Note that as the true demand type is common knowledge, there is no need on the side of suppliers to update their beliefs about whether $\phi = l$ or $\phi = h$. For this information scenario, we first characterize the equilibria under RFQ^n and RFQ^r and analyze the equilibrium decisions and profits/costs of supply chain firms.

4.1. The Equilibrium Characterization

Recall that there are three decision stages in the game, namely, RFQ, pricing, and ordering stages. Since the last two stages are common to both types of RFQs, we can combine the analyses of the pricing and ordering stages and characterize the Stackelberg equilibria using backward induction, starting from the buyer's optimal (cost-minimizing) order allocation stage. Let $TC_B^\phi(p_U, p_R)$ be the expected minimum total cost function for a ϕ-type buyer given that suppliers U and R offer p_U and p_R, respectively. Then, this cost function can be defined as follows:

$$\begin{aligned} TC_B^\phi(p_U,p_R) = \min_{q_U\ge 0, q_R\ge 0} \; & \beta^\phi \cdot (Q - q_U - q_R)^+ \\ & + (\alpha p_U + (1-\alpha)\beta^\phi)\cdot q_U + p_R \cdot q_R \\ \text{s.t.} \quad & \\ 0 \le q_U + q_R \le Q, & \quad \text{if } RFQ^n \text{ is offered;} \\ I_F(1-\underline{\omega}) \le q_U + q_R \le I_F(1+\bar{\omega}), & \quad \text{if } RFQ^r \text{ is offered;} \end{aligned} \tag{3}$$

where the first term in the objective function represents the buyer's expected cost of using his insourcing option, and the second and third terms represent the buyer's expected cost of buying from suppliers U and R, respectively (refer to Appendix for the derivation of these expressions). Note that since the buyer's order allocation is a linear problem, depending on the coefficients of q_U and q_R, one of the constraints will always be binding at the optimal solution. Once the optimal order allocation decision is characterized, we substitute it into the suppliers profit functions and then solve the simultaneous price game between suppliers U and R. Note that the price competition between suppliers U and R always results in one of the suppliers charging her marginal cost (c_R for supplier R and $\frac{c_U}{\alpha}$ for supplier U) and the other supplier charging a price such that the expected cost for the buyer in buying from her is infinitesimally less than his expected

Table 2 Equilibrium Characterization under Symmetric Information Setting

		Low demand $\beta^{\phi} \le \gamma$	High demand $\beta^{\phi} > \gamma$
RFQs	RFQ^n	any $I_F \in [0,Q]$	any $I_F \in [0,Q]$
	RFQ^r	$I_F(1-\underline{\omega}) = 0$, $I_F(1+\bar{\omega}) = Q$	$I_F(1-\underline{\omega}) = 0$, $I_F(1+\bar{\omega}) = Q$
Prices	$p_R^{\phi*}$, $p_U^{\phi*}$	c_R, $\left[\frac{c_U}{\alpha} + \frac{(1-\alpha)(\gamma-\beta^{\phi})}{\alpha}\right]^{(-)}$	$[c_R + (1-\alpha)(\beta^{\phi}-\gamma)]^{(-)}$, $\frac{c_U}{\alpha}$
Order allocation	$q_R^{\phi*}$,$q_U^{\phi*}$	$0,Q$	$Q,0$
Suppliers' profits	$\pi_R^{\phi*}$, $\pi_U^{\phi*}$	0, $(c_R - (1-\alpha)\beta^{\phi} - c_U)Q$	$(c_U + (1-\alpha)\beta^{\phi} - c_R)Q$, c_U
Buyer's cost	$TC_B^{\phi*}$	$c_R Q$	$(c_U + (1-\alpha)\beta^{\phi})Q$

cost of buying from her opponent. We delegate the detailed analysis to the Appendix and provide the overall equilibrium characterization in Proposition 1:

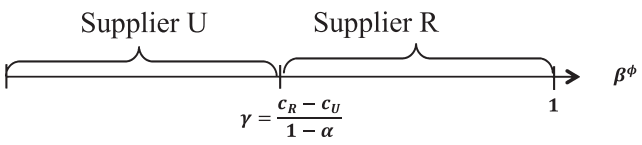
PROPOSITION 1. *Let* $\gamma = \frac{c_R - c_U}{1-\alpha}$. *Then, under symmetric information equilibrium, the buyer orders from supplier U if* $\beta^{\phi} \le \gamma$, *otherwise, he orders from supplier R. The complete characterization of equilibrium decisions, profits and costs for the supply chain parties under symmetric information are provided in Table* 2 *for both* RFQ^n *and* RFQ^r.

Note that when the demand is expected to be low, supplier U becomes the sole supplier for the buyer and sets a price $p_U^{\phi*}$ such that the buyer's expected cost of buying from her $(= \alpha p_U^{\phi*} + (1-\alpha)\beta^{\phi})$ is infinitesimally less than supplier R's marginal c_R. Letting $\alpha p_U^{\phi*} + (1-\alpha)\beta^{\phi} = c_R$ and solving for $p_U^{\phi*}$ would then yield the equilibrium price for supplier U. As shown in Table 2, the equilibrium price for supplier U can be expressed in terms of his marginal cost (c_U/α) plus a markup term that depends on the demand uncertainty, and the suppliers' cost and risk parameters. In section 4.2, we explain these terms in more detail. Similarly, when the demand is expected to be high, supplier R can then set the price $p_R^{\phi*}$ in a similar fashion and become the sole supplier for the buyer.

4.2. The Impact on the Equilibrium Decisions under Symmetric Information

First of all, as shown in Figure 3, the buyer prefers suppliers U and R for low and high values of β^{ϕ}, respectively; the intuition behind this comes from the comparison between relative overage and underage costs[11] associated with suppliers U and R.

Figure 3 Equilibrium order Allocation under Symmetric Information Setting



- By ordering one more unit from supplier U as opposed to supplier R, the buyer's expected cost changes by $(1-\alpha + \alpha p_U - p_R)$ if demand is realized and by $(\alpha p_U - p_R)$ if it is not. The relative comparison between these expressions suggests that ordering from supplier U instead of from R would increase[12] the buyer's expected cost more when the demand is realized than when it is not. In other words, if the demand is more likely to be realized, the buyer would reduce his underage cost by ordering more from supplier R since she is more secure than supplier U. On the other hand, if demand is less likely to be realized, the buyer would prefer to order more from supplier U to lower his overage cost. As characterized in proposition 1, this implies a threshold $(= \gamma)$ for β^{ϕ}, above which the buyer opts for ordering from supplier R (see region "supplier R" in Figure 3), and below which he opts for sourcing from supplier U (see regions "supplier U" in Figure 3).
- Further analysis of this threshold $(\gamma = \frac{c_R - c_U}{1-\alpha})$ yields a comparative metric for supplier U that shows how she compares with supplier R in terms of cost $(c_R - c_U)$ and yield risk $(1-\alpha)$. This metric implies that the higher the cost differential and/or the lesser the yield risk differential is between the suppliers, the higher the chances are for supplier U to receive the order allocation from the buyer.

The above findings also imply that the equilibrium prices charged by the suppliers depend on the extent of demand faced by the ϕ-type buyer (as measured by β^{ϕ}) and the cost-to-risk ratio of supplier U with respect to supplier R (as measured by γ). In general, lower values of β^{ϕ} and/or higher values of γ make supplier U more preferable for the buyer, which in turn enables her to gain more pricing power against R and consequently increase her price $p_U^{\phi*}$.

Finally, the equilibrium profits and costs for the channel partners are exactly the same under both RFQ^n and RFQ^r. So, when all the supply chain partners have access to the same demand forecast information, a specific RFQ type does not play a sig-

nificant role. However, as we will see in the next section, when there is information asymmetry among the supply chain parties about the demand risks facing the buyer, both the sourcing and pricing strategies change significantly with the type of RFQ offered by the buyer.

5. Equilibrium Analysis under an Asymmetric Information Scenario

In this section, we analyze the equilibrium for the case where the true demand type is known only to the buyer and the other two suppliers face information asymmetry about the true demand type. Recall that in the presence of incomplete information, the suppliers need to rely on their updated beliefs about the probability of demand being of *h*- or *l*-type (denoted by $\hat{r}^h$ and $\hat{r}^l$, respectively, where $\hat{r}^h + \hat{r}^l = 1$) to make pricing decisions. We analyze RFQ^n and RFQ^r in sections 5.1 and 5.2, respectively.

5.1. The Equilibrium Characterization of RFQ^n under Asymmetric Information

Following a similar approach as the one in section 4, we can characterize the equilibrium using backward induction, starting from the buyer's optimal (cost-minimizing) order allocation stage, given that suppliers U and R offer p_U and p_R, respectively. Note that since the buyer solves the same order allocation problem as the one under symmetric information setting (3), the equilibrium order allocation will be exactly same. However, there is a significant difference between symmetric and asymmetric information settings. Indeed, under the asymmetric information setting, there are two different allocation policies associated with two demand types, and suppliers do not know which allocation policy will eventually be used by the buyer unless he signals his true demand information ($\phi = h$ or $\phi = l$) via an RFQ^n. In the next proposition, we show that the buyer cannot credibly signal his true demand type under RFQ^n:

Proposition 2. *Under RFQ^n, the only equilibrium in an asymmetric information scenario is of pooling type, where both the h-type and the l-type buyer submit the same initial forecast, i.e., $I_F^h = I_F^l$.*

The above proposition implies that credible forecast sharing is not sustainable in equilibrium under RFQ^n because either *l*- or *h*-type buyer can always decrease his expected cost by imitating the other type if they offer different I_F. Therefore, in equilibrium, the suppliers' updated beliefs will be the same as their prior beliefs, i.e., $\hat{r}^\phi = r^\phi$ for both $\phi \in \{l,h\}$. In the following proposition, we characterize the equilibrium of the pricing game between suppliers U and R in the presence of incomplete information about the true demand type:

Proposition 3. *Under RFQ^n, the suppliers' equilibrium pricing strategies and expected profits and the buyer's expected cost are characterized in Table 3 for all the cases shown in Figure 4.*

A closer examination of Table 3 and Proposition 3 reveals that the equilibrium characterization crucially depends on the suppliers' *a priori* beliefs about the probability distribution of the true demand type, i.e., r^h and r^l. To see why this is so, recall that in the symmetric information setting, when one of the suppliers undercuts her opponent's price, she does so by taking into account the true demand type of the buyer. However, as shown in proposition 3, the suppliers cannot credibly infer the demand type, which forces each supplier to decide between the following two alternatives: (i) either she undercuts the opponent's price for both *l*- and *h*-type demand scenarios (*mass-market strategy*), (ii) or she does it for only one of the demand scenarios (*niche-market strategy*). The first alternative (mass-market strategy) enables her to win the order allocation from the buyer under both demand scenarios, $\phi \in \{l,h\}$. But to be able to do so, she needs to decrease her price quite significantly. On the other hand, the second alternative (niche-market strategy) allows her to charge a higher price but leaves her with a probabilistic order allocation, the likelihood of which depends on *a priori* beliefs for the targeted niche-market demand scenario (*h*-type demand for supplier R and *l*-type demand for supplier U).

If the likelihood of a niche-market demand scenario is very low (the case represented by C_U^1, and C_R^1 in Table 3), both suppliers would opt for mass-market strategy, which results in *head-to-head* price competition between them. As we discuss later (section 6), in such cases, the buyer would benefit from such a mass-market strategy because the intensive price competition resulting from the head-to-head price competition between both suppliers helps equilibrium prices to decrease. On the other hand, when both suppliers can extract positive profits from their respective niche-market demand scenarios (i.e., cases C_U^2, C_R^2, and C_{UR}), instead of engaging in head-to-head competition for the same demand scenario(s), they *split* the market among each other. Specifically, in this case, supplier U undercuts supplier R for the *l*-type demand and supplier R undercuts supplier U for the *h*-type demand. As we discuss in section 6, this *weakens* the strength of price competition between the suppliers, which enables them to increase their prices.

Table 3 Equilibrium characterization of *RFQ*^{n} under asymmetric information (for the cases C_U, C_R, and C_{UR})

Regions and Conditions		$(\beta^l,\beta^h) \in C_U$: $\beta^l \le \beta^h \le \gamma = \frac{c_R - c_U}{1-\alpha}$		$(\beta^l,\beta^h) \in C_{UR}$: $\beta^l \le \gamma \le \beta^h$	$(\beta^l,\beta^h) \in C_R$: $\gamma \le \beta^l \le \beta^h$	
		C_U^1: $r^h \ge \frac{\beta^h - \beta^l}{\gamma - \beta^l}$	C_U^2: $r^h < \frac{\beta^h - \beta^l}{\gamma - \beta^l}$	for all $r^h \in [0,1]$	C_R^2: $r^h > \frac{\beta^l - \gamma}{\beta^h - \gamma}$	C_R^1: $r^h \le \frac{\beta^l - \gamma}{\beta^h - \gamma}$
Prices	p_R^*	c_R	$\sim F_R(p_R) = \frac{1}{r^h}\left(1 - \frac{r^l(\overline{p}_R - \bar{\alpha}\beta^h - c_U)}{p_R - \bar{\alpha}\beta^h - c_U}\right)$ for $p_R \in \left[\underline{p}_R, \overline{p}_R\right]$			$\left[c_U + \bar{\alpha}\beta^l\right]^{(-)}$
	p_U^*	$\left[\frac{c_R - \bar{\alpha}\beta^h}{\alpha}\right]^{(-)}$	$\sim F_U(p_U) = 1 - \frac{\alpha\underline{p}_U + \bar{\alpha}\beta^h - c_R}{\alpha p_U + \bar{\alpha}\beta^h - c_R}$ for $p_U \in \left[\underline{p}_U, \overline{p}_U\right]$			$\frac{c_U}{\alpha}$
Alloc	(q_U^l, q_R^l); (q_U^h, q_R^h)	$(Q,0)$; $(Q,0)$	$\begin{cases}(Q,0);(Q,0) & \text{if } \alpha p_U^* + \bar{\alpha}\beta^h \le p_R^* \\ (Q,0);(0,Q) & \text{if } \alpha p_U^* + \bar{\alpha}\beta^h > p_R^*\end{cases}$			$(0,Q)$; $(0,Q)$
Profits	π_R^*	0	$r^h(\underline{p}_R - c_R)Q$			$(c_U + \bar{\alpha}\beta^l - c_R)Q$
	π_U^*	$(c_R - \bar{\alpha}\beta^h - c_U)Q$	$r^l(\alpha\overline{p}_U - c_U)Q$			0
Cost	TC_B^{l*}	$(c_R - \bar{\alpha}(\beta^h - \beta^l))Q$	$\underline{p}^l + (\underline{p}^l + \bar{\alpha}\beta^h - \bar{\alpha}\beta^l - c_R)\ln\left(\frac{\overline{p}^l + \bar{\alpha}\beta^h - \bar{\alpha}\beta^l - c_R}{\underline{p}^l + \bar{\alpha}\beta^h - \bar{\alpha}\beta^l - c_R}\right)$			$(c_U + \bar{\alpha}\beta^l)Q$
	TC_B^{h*}	$c_R Q$	$\underline{p}^h + \frac{r^l}{r^h}(\underline{p}^h - c_R)\left[\frac{\overline{p}^h - c_R}{c_R - \bar{\alpha}\beta^h - c_U}\ln\left(\frac{\overline{p}^h - c_R}{\underline{p}^h - c_R}\right) + \frac{\overline{p}^h - \bar{\alpha}\beta^h - c_U}{c_U + \bar{\alpha}\beta^h - c_R}\ln\left(\frac{\overline{p}^h - \bar{\alpha}\beta^h - c_U}{\underline{p}^h - \bar{\alpha}\beta^h - c_U}\right)\right]$			$(c_U + \bar{\alpha}\beta^l)Q$

Note: Closed-from expressions for $\bar{p}^\phi$, $\underline{p}^\phi$ for $\phi \in \{l,h\}$, and $\bar{p}_U$, $\underline{p}_U$, $\bar{p}_R$ and $\underline{p}_R$ are provided in the Appendix.

Such a scenario indeed hurts the buyer and therefore motivates him to search for a signal to credibly communicate his demand information to the suppliers, which would eventually make them engage in head-to-head price competition again.

5.2. The Equilibrium Characterization for RFQ^r under Asymmetric Information

In this section, we consider the impact of *RFQ*^{r} on equilibrium decisions as well as on the credibility of forecast information sharing between supply chain partners.

PROPOSITION 4. *The type of equilibrium as well as decisions, profits, and cost under an RFQ*^{r} *are characterized for all the cases shown in Figure* 4 *as follows:*

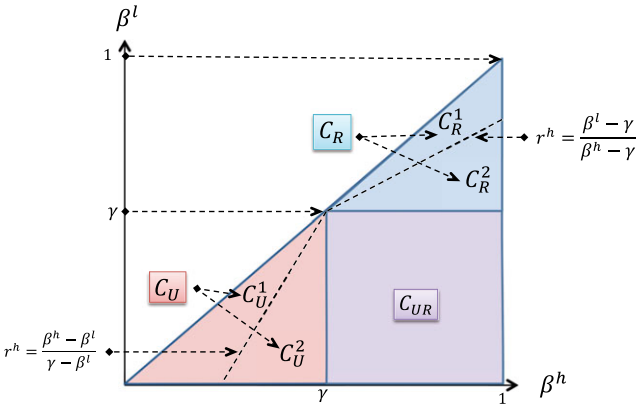
- *The equilibrium under an RFQ*^{r} *is of separating type if and only if* $(\beta^l,\beta^h) \in C_U$ *or* $(\beta^l,\beta^h) \in C_R$. *The equilibrium allocation, prices, updated beliefs, and expected profits/costs for the supply chain parties for these two regions are provided in Table 4.*
- *For the remaining cases (for case* C_{UR}*), the equilibrium under an RFQ*^{r} *is of pooling type, i.e.,* $\hat{r}^\phi = r^\phi$ *for* $\phi = l,h$. *The equilibrium allocation, prices, and expected profits/costs for the supply chain parties for the pooling cases are the same as the ones characterized in Table 3 (for case* C_{UR}*).*

The main takeaway from proposition 2 is that a buyer who has private information about his true demand type can convey this information in a credible way via an *RFQ*^{r}. Below, we discuss first *how* and then *when* an *RFQ*^{r} can be used as a credible signal.

The answer to the *how* question depends on whether an *l*- or *h*-type buyer has an incentive to imitate the *RFQ*^{r} offered by the other type. There are two cases: (i) when the extent of demand faced by both types of buyer is high, i.e., $(\beta^l,\beta^h) \in C_R$, and (ii) when it is low, i.e., $(\beta^l,\beta^h) \in C_U$. We focus on the former case to illustrate the intuition.[13] Recall from Table 2 in section 4 that in C_R, when the suppliers compete for the same demand type, the resulting equilibrium price offered for the *l*-type buyer is less than that for the *h*-type buyer. This implies that in the presence of asymmetric information, an *h*-type buyer always has an incentive to imitate the RFQ offered by an *l*-type buyer. So, in order for the *l*-type to separate himself from the *h*-type, he needs to design an *RFQ*^{r} that is too costly for an *h*-type to mimic. This can be accomplished by the *l*-type buyer via a *costly* self-imposed upper bound on the amount of products that he will order in the future.

We can formalize this as follows: let $(I_F^{l*}, \underline{\omega}^{l*}, \bar{\omega}^{l*})$ and $(I_F^{h*}, \underline{\omega}^{h*}, \bar{\omega}^{h*})$ be two different *RFQ*^{r}s offered by the *l*- and the *h*-type buyers, respectively. In response to this, both suppliers update their *a priori* beliefs accordingly and charge (p_U^{l*}, p_R^{l*}) and (p_U^{h*}, p_R^{h*}) for *l*- and *h*-type buyers, respectively. In order for these

Figure 4 Equilibrium Characterization Cases for *RFQ*^{n} and *RFQ*^{r} under Asymmetric Information



Note: Refer to Table 3 for equilibrium characterization of RFQ^n for the cases C_U, C_R, and C_{UR}, and RFQ^r for C_{UR}, and refer to Table 4 for equilibrium characterization of RFQ^r for the cases C_U and C_R.

Table 4 Equilibrium characterization of *RFQ*^{r} under Asymmetric Information (for the cases C_U and C_R).

Regions		$(\beta^l,\beta^h) \in C_R$: $\gamma \le \beta^l \le \beta^h$	$(\beta^l,\beta^h) \in C_U$: $\beta^l \le \beta^h \le \gamma$
RFQs	$(I_F^\phi, \underline{\omega}^\phi, \bar{\omega}^\phi)$	any $(I_F^\phi, \underline{\omega}^\phi, \bar{\omega}^\phi)$ s.t. $I_F^l(1+\bar{\omega}^\phi) = \begin{cases} Q & \text{if } \phi = h \\ \frac{1+(\beta^l - c_U/\alpha)/(\beta^h-\beta^l)}{1/\alpha+(\beta^l-c_U/\alpha)/(\beta^h-\beta^l)} Q & \text{if } \phi = l \end{cases}$	any $(I_F^\phi, \underline{\omega}^\phi, \bar{\omega}^\phi)$ s.t. $I_F^l(1+\underline{\omega}^\phi) = \begin{cases} \frac{c_R-\beta^l\bar{\alpha}}{c_R-\beta^h\bar{\alpha}} Q & \text{if } \phi = h \\ Q & \text{if } \phi = l \end{cases}$
Updated beliefs	$(\hat{r}^h, \hat{r}^l)$	$\begin{cases} (1,0) & \text{if } I_F(1+\bar{\omega}) > I_F^l(1+\bar{\omega}^l) \\ (0,1) & \text{if } I_F(1+\bar{\omega}) \le I_F^l(1+\bar{\omega}^l) \end{cases}$	$\begin{cases} (1,0) & \text{if } I_F(1-\underline{\omega}) \ge I_F^h(1-\underline{\omega}^h) \\ (0,1) & \text{if } I_F(1-\underline{\omega}) < I_F^h(1-\underline{\omega}^h) \end{cases}$
Prices	p_U^*, p_R^*	$\begin{cases} \frac{c_U}{\alpha}, \left[c_U + \bar{\alpha}\beta^h\right]^{(-)} & \text{if } I_F(1+\bar{\omega}) > I_F^l(1+\bar{\omega}^l) \\ \frac{c_U}{\alpha}, \left[c_U + \bar{\alpha}\beta^l\right]^{(-)} & \text{if } I_F(1+\bar{\omega}) \le I_F^l(1+\bar{\omega}^l) \end{cases}$	$\begin{cases} \left[\frac{c_R-\bar{\alpha}\beta^h}{\alpha}\right]^{(-)}, c_R & \text{if } I_F(1-\underline{\omega}) \ge I_F^h(1-\underline{\omega}^h) \\ \left[\frac{c_R-\bar{\alpha}\beta^l}{\alpha}\right]^{(-)}, c_R & \text{if } I_F(1-\underline{\omega}) < I_F^h(1-\underline{\omega}^h) \end{cases}$
Alloc	$q_U^{\phi*}$, $q_R^{\phi*}$	$0, I_F^\phi(1+\bar{\omega}^\phi)$	$I_F^\phi(1-\underline{\omega}^\phi), 0$
Suppliers' profits	π_R^*, π_U^*	$\sum_{\phi=l,h} r^\phi (c_U + \bar{\alpha}\beta^\phi - c_R) I_F^\phi(1+\bar{\omega}^\phi), 0$	$0, \sum_{\phi=l,h} r^\phi (c_R - \bar{\alpha}\beta^\phi - c_U) I_F^\phi(1-\underline{\omega}^\phi)$
Buyer's cost	TC_B^{h*}	$(c_U + \bar{\alpha}\beta^h) I_F^h(1-\bar{\omega}^h)$	$c_R I_F^h(1-\underline{\omega}^h) + \beta^h\bar{\alpha}(I_F^h(1-\underline{\omega}^h) - Q)$
	TC_B^{l*}	$\beta^l(Q - I_F^l(1+\bar{\omega}^l)) + (c_U + \bar{\alpha}\beta^l) I_F^l(1+\bar{\omega}^l)$	$c_R I_F^l(1+\underline{\omega}^l)$

Refer to Table 3 for the equilibrium characterization of *RFQ*^{r} for the case C_{UR}, i.e., $\beta^l \le \gamma \le \beta^h$.

RFQs to be *credible* in equilibrium, and updated beliefs to be consistent with the buyer's decisions, equilibrium RFQ^r and prices should satisfy the following incentive compatibility (IC) constraints:

$$\text{IC constraint for } h-\text{type buyer}: TC_B^h(p_U^{h*}, p_R^{h*}|I_F^{h*}, \underline{\omega}^{h*}, \bar{\omega}^{h*}) \le TC_B^h(p_U^{l*}, p_R^{l*}|I_F^{l*}, \underline{\omega}^{l*}, \bar{\omega}^{l*})$$
$$\text{IC constraint for } l-\text{type buyer}: TC_B^l(p_U^{l*}, p_R^{l*}|I_F^{l*}, \underline{\omega}^{l*}, \bar{\omega}^{l*}) \le TC_B^{l*}(p_U^{h*}, p_R^{h*}|I_F^{h*}, \underline{\omega}^{h*}, \bar{\omega}^{h*})$$

where $TC_B^{\phi}(p_U^{\phi'*}, p_R^{\phi'*}|I_F^{\phi'*}, \underline{\omega}^{\phi'*}, \bar{\omega}^{\phi'*})$ denotes the expected total cost (defined in Equation 3) incurred by a ϕ-type buyer provided that he offers an RFQ^r designed by a ϕ'-type buyer, where $\phi \in \{l,h\}$, $\phi' \in \{l,h\}$, and $\phi \neq \phi'$. In other words, the above two conditions ensure that neither the *l*- nor the *h*-type buyer has any incentive to deviate from his respective RFQ^r even if doing so would lead to lower prices. Note that there can be multiple solutions to the above conditions. To eliminate the multiplicity of the equilibria, we employ the *intuitive criterion*, a commonly used equilibrium refinement concept developed for the signalling games (Cho and Kreps 1987). Indeed, this refinement enables us to determine a *unique* equilibrium RFQ^r that is both incentive compatible and least costly from both the *l*- and *h*-type buyers' perspectives.

The answer to the *when* question depends on two things: (i) the impact of offering a separating RFQ^r on the strength of price competition and (ii) the cost of sustaining a separating equilibrium. Recall that, as shown in section 5.1, the lack of credibility may weaken the price competition, which can be restored via an RFQ^r offering. However, to make the RFQ^r offering credible in the eyes of suppliers, the party offering RFQ^r has to burden a cost, the *signalling cost*, which increases in the degree of informational asymmetry across the supply chain (measured by $\beta^h - \beta^l$). Hence, if the degree of information asymmetry is relatively low (i.e., C_U and C_R), the signalling cost would also be low, and the separating RFQs can be sustained in equilibrium. On the other hand, if the degree of information asymmetry is very high (i.e., C_{UR}), then, the buyer would find it too costly to sustain separating RFQs even if they strengthen the degree of competition between the suppliers.

6. The Impact of RFQ^n and RFQ^r

In this section, we first compare the effects of RFQ^n and RFQ^r on the equilibrium decisions, and subsequently, on the equilibrium costs and profits.

6.1. Impact On Equilibrium Decisions

The following proposition compares the impact of an RFQ^r on the equilibrium channel decisions with respect to an RFQ^n:

PROPOSITION 5. *The impact of an RFQ^r on equilibrium order quantities and prices with respect to an RFQ^n under asymmetric information setting is characterized as follows:*

- *Equilibrium order quantities decrease in C_U and increase in C_R under an RFQ^r, with respect to an RFQ^n.*
- *Equilibrium prices seen by both l- and h-type buyers increase in cases C_U^1 and C_R^1, and decrease in the cases C_R^2 and C_U^2 (except for l-type buyer in C_U^2) under an RFQ^n, with respect to an RFQ^n.*

As characterized in proposition 2, in order for the buyer to use an RFQ^r as a credible signal, he needs to impose either a lower or an upper bound on the order quantity. These self-imposing constraints will increase and decrease the equilibrium order quantity in regions C_R and C_U, respectively. Note that the relative cost-to-risk ratio γ between suppliers U and R also plays an important role in the format of RFQ^r offered by the buyer. Specifically, the higher the cost-to-risk ratio of supplier U with respect to R (i.e., higher values of γ), the more likely the buyer is to offer an RFQ^r with upper bounds on his order quantity to incentivize supplier U to engage in head-to-head competition with supplier R. This would in turn result in a lower order quantity for the buyer under an RFQ^r with respect to RFQ^n.

With regards to the effects of RFQ decisions on equilibrium prices, we need to take into account whether RFQ^n and RFQ^r lead suppliers U and R to (i) engage in head-to-head competition for the same demand scenario(s) or split the market among each other, and (ii) opt for mass- or niche-market strategies. Results are summarized in Table 5.

Table 5 shows that an RFQ^n decreases the strength of price competition between suppliers from one extreme (where they engage in head-to-head competition for mass market) to another (where each supplier targets at her respective niche-market demand scenario), whereas an RFQ^r keeps the strength of price competition in the middle (where suppliers U and R engage in head-to-head competition for the same niche market). With respect to RFQ^n, this results in a higher equilibrium price under RFQ^r for regions C_U^1 and C_R^1, and a lower equilibrium price under RFQ^r for regions C_U^2 and C_R^2 (except for *l*-type buyer in C_U^2).

We also analyze the impact of *a priori* beliefs on the equilibrium prices charged by the suppliers under both RFQ^n and RFQ^r. Note that, as shown in Figure 4, the regions C_U^2 and C_R^2, where suppliers U and R each

Table 5 Impact of RFQ^n and RFQ^r on the Strength of Price Competition

(Competition weakens and price increases) ⇒			
		Mass-market strategy	**Niche-market strategy**
(Competition weakens / Price increases) ⇓	**Head-to-head competition**	RFQ^n in C_U^1 and C_R^1	RFQ^r
	Split demand scenarios	Not applicable	RFQ^n in C_U^2 and C_R^2

target at their respective niche demand scenarios under an RFQ^n, increase in size in r^l and r^h, respectively. This follows intuition because in C_U^2 (resp., C_R^2), the profitability of niche demand scenario for supplier U (resp., R) increases in r^l (resp., r^h). This in turn increases the value of an RFQ^r from the buyer's perspective because it enables the suppliers to compete for the same demand scenario, resulting in lower prices for the buyer.

6.2. On Equilibrium Profits/Costs

Using the results of the previous section, the following proposition establishes the impact of an RFQ^r on the equilibrium costs and profits of the channel partners with respect to an RFQ^n:

PROPOSITION 6. *The comparison between the channel partners' profits and costs under RFQ^n and RFQ^r in an asymmetric information setting is characterized in Figure 5.*

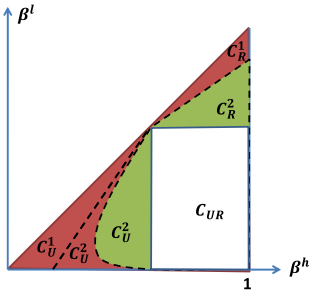
The main takeaway from the above proposition is that ultimately it is the degree of information asymmetry (i.e., $\beta^h - \beta^l$) that causes differences in expected costs and profits of channel partners between RFQ^n and RFQ^r. We elaborate on this by considering the following three cases:

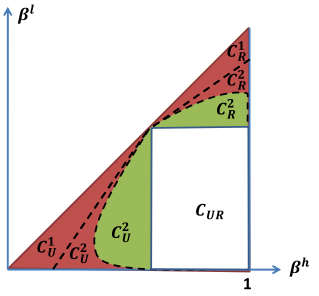
- When the degree of information asymmetry is low (i.e., regions C_U^1 and C_R^1): Note that the buyer is always worse off with an RFQ^r. This is because under an RFQ^r, the buyer faces two setbacks: an additional signalling cost and rising supply prices due to a weaker price competition between the suppliers. To sum, when the degree of information asymmetry is low, the buyer does not gain any benefit from the information visibility. On the other hand, from the suppliers' perspectives, they are better off with the buyer offering an RFQ^r because it allows them to increase their prices.
- When the degree of information asymmetry is medium (i.e., regions C_U^2 and C_R^2): As the extent of demand facing *l*- and *h*-type buyers gets dissimilar, it becomes too costly for each supplier to offer a price low enough to cover all the demand scenarios. Consequently, the suppliers have a greater incentive to split the demand scenarios among each other, which in turn weakens the competition.
- This creates a trade-off situation for the buyer. Namely, the buyer can strengthen the competition by offering an RFQ^r, but he has to incur a cost to make it credible. Based on the cost–benefit analysis, Figure 5 indicates that the buyer gains more from an RFQ^r than he loses as the degree of information asymmetry grows. On the other hand, from the suppliers' perspective, Figure 5 shows that suppliers are generally better off with an RFQ^n and prefer an RFQ^r only when the degree of information asymmetry is sufficiently low. There are two reasons for this. First, RFQ^n allows the suppliers to increase their price offerings. Second, it may also increase the order quantity for them. For example, consider region C_R^2. In this region, under an RFQ^r, supplier U will always be undercut by supplier R for all demand scenarios. However, under an RFQ^n, supplier R focuses on the *h*-type demand and leaves the other demand scenario (*l*-type) for supplier U to win.
- Finally, when the degree of information asymmetry is high (i.e., region C_{UR}): As shown in proposition 2, since it is too costly for the buyer to credibly convey his demand information to the suppliers, the only equilibrium that is sustainable in C_{UR} is of pooling type, which implies that an RFQ^r has no impact either on the buyer's expected cost or on the suppliers' expected profits.

7. Conclusion

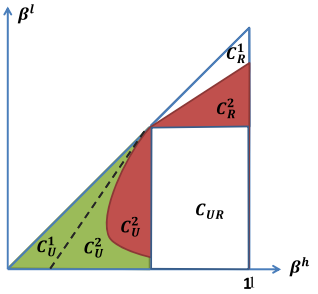
In this study, we analyze how and when a buyer can credibly share his forecast information with his upstream suppliers, and how it impacts the equilibrium decisions and the profits/costs of channel partners. To address these questions, we develop a decentralized supply chain model with a buyer facing private demand information and two heterogeneous suppliers competing for the buyer's order quantity. We analyze two RFQ types, non-restrictive (RFQ^n) and restrictive (RFQ^r). In each case, the buyer obtains a private signal (forecast) for the demand and decides on the RFQ parameters. Then, the suppliers update

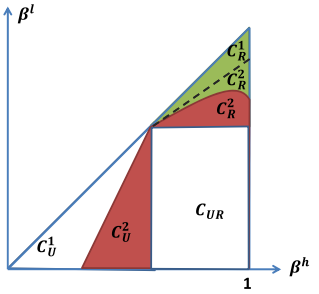
Figure 5 Effects of RFQ^n vs. RFQ^r on Supply Chain Partners' Profit/Costs in Asymmetric Information Setting





Regions		*h*-type Buyer's Cost	*l*-type Buyer's Cost
C_U^1:	$(\beta^l, \beta^h) \in C_U$ and $r^l \le 1/\zeta_U$	$RFQ^n > RFQ^r$	$RFQ^n > RFQ^r$
C_U^2:	$(\beta^l, \beta^h) \in C_U$ and $r^l > 1/\zeta_U$	$RFQ^n \ge RFQ^r$ if $\Delta_1^h \ge 0$; $RFQ^n < RFQ^r$ o.w.	$RFQ^n \ge RFQ^r$ if $\Delta_1^l \ge 0$; $RFQ^n < RFQ^r$
C_R^1:	$(\beta^l, \beta^h) \in C_R$ and $r^h \le \zeta_R$	$RFQ^n > RFQ^r$	$RFQ^n > RFQ^r$
C_R^2:	$(\beta^l, \beta^h) \in C_R$ and $r^h > \zeta_R$	$RFQ^n < RFQ^r$	$RFQ^n \ge RFQ^r$ if $\Delta_2^l \ge 0$; $RFQ^n < RFQ^r$
C_{UR}:	$(\beta^l, \beta^h) \in C_{UR}$	$RFQ^n = RFQ^r$	$RFQ^n = RFQ^r$





Regions		Supplier U's profit	Supplier R's profit
C_U^1:	$(\beta^l, \beta^h) \in C_U$ and $r^l \le 1/\zeta_U$	$RFQ^n < RFQ^r$	$RFQ^n = RFQ^r$
C_U^2:	$(\beta^l, \beta^h) \in C_U$ and $r^l > 1/\zeta_U$	$RFQ^n \ge RFQ^r$ if $\Delta_U \ge 0$; $RFQ^n < RFQ^r$ o.w.	$RFQ^n \ge RFQ^r$
C_R^1:	$(\beta^l, \beta^h) \in C_R$ and $r^h \le \zeta_R$	$RFQ^n = RFQ^r$	$RFQ^n < RFQ^r$
C_R^2:	$(\beta^l, \beta^h) \in C_R$ and $r^h > \zeta_R$	$RFQ^n \ge RFQ^r$	$RFQ^n \ge RFQ^r$ if $\Delta_R \ge 0$; $RFQ^n < RFQ^r$
C_{UR}:	$(\beta^l, \beta^h) \in C_{UR}$	$RFQ^n = RFQ^r$	$RFQ^n = RFQ^r$

Note: The different colored regions in the above figure denote the following preference between RFQ^n and RFQ^r for each supply chain partner: green (light shaded) regions - prefers RFQ^r over RFQ^n; red (dark shaded) regions - prefers RFQ^n over RFQ^r; and, white regions - indifferent between RFQ^n and RFQ^r. Δ_i^ϕ (for ϕ = l, h for i = 1, and ϕ = l, and i = 2), Δ_U and Δ_R are characterized in Appendix.

their beliefs on the true forecast type, and compete on prices to earn the order from the buyer. Once the supply and demand risks are realized, and orders are shipped, the buyer clears the supply–demand mismatch, if there is any.

We first characterize the equilibrium under symmetric information and show that both RFQ^n and RFQ^r lead to same equilibrium decisions, profits and costs of the channel partners. The choice of sourcing strategy depends on the amount of forecasted demand. Specifically, the unreliable supplier (U) is preferred for low forecasted demand scenarios, while the reliable supplier (R) is preferred for high forecasted demand scenarios. The analyses of RFQ^n and RFQ^r under the asymmetric information setting bring new twists to every aspect of the problem, especially to the credibility of the forecast information and the strength of price competition at the supplier level. First, the fact that the suppliers hold incomplete information about the demand leads to an uncertainty on their side regarding what price to charge, which, in turn, generates an incentive for the buyer to distort his forecast information, and invalidates the credibility of an RFQ^n. We show that this distortion can be fixed, albeit at a cost, via an RFQ^r, as long as the degree of information asymmetry between channel partners is not too high.

Second, in the presence of incomplete information, the suppliers decide on their prices solely based on their *a priori* beliefs on the true demand scenario, and choose (i) whether to opt for mass- or niche-market demand scenario, and (ii) whether to engage in *head-to-head competition* for the same demand scenario or split the demand scenarios between themselves. Depending on the choices made by the suppliers, the strength of price competition (resp., equilibrium price) decreases (resp., increases) from one extreme to another as the suppliers move from mass- to niche-market, and from head-to-head competition for the same market to splitting them among each other. Our results show that an RFQ^n pushes the degree of price

competition to the extremes at either end, whereas an RFQ^r keeps it in the middle. Finally, a comparison of the cost/profit implications of the above two effects on the payoffs enables us to evaluate when an RFQ^r is more preferable option (and when it is not) for each channel partner.

The above-mentioned results give some managerial insights on the role of credible information sharing in competitive supply chains:

First, the degree of informational asymmetry between the buyer and the suppliers has to be in the medium range (i.e., medium level of $\beta^h - \beta^h$) in order for the buyer to justify the cost associated with providing demand visibility via an RFQ^r. This suggests that the restrictions on the order quantity can be used as an effective tool in the later stages of procurement relations between the buyer and the suppliers, when they have already somewhat established a relationship. This could offer an explanation of why Sun MicroSystems initially started with no commitments and then gradually moved to more restrictive procurement mechanisms with its suppliers for various workstation/server components.

Next, our results suggest that the buyer may need to customize the type of restriction that he provides with RFQ^r depending on the level of reliability of the supplier he is interacting with. Specifically, if the buyer is interacting with an unreliable supplier (i.e., supplier U), then he would need to impose a lower bound on the amount of products to be purchased from the supplier. Indeed, minimum capacity reservations, a common contractual form used in the semiconductor industry, where producers' yields are not too high due to the high degree of production risks, are similar to the agreements with a lower bound on the order quantity. On the other hand, if the buyer is interacting with a reliable supplier (i.e., supplier R), then he would instead offer an upper bound on the amount of products to be purchased. Indeed, as indicated in Farlow et al. (1996b), Sun MicroSystems used a specific form with only an upper bound constraint for the suppliers of relatively low risk items such as keyboards and monitors.

Finally, even though we consider specific values for the cost of internal production and the salvage value (modelled through $c_I = 1$ and $c_S = 0$, respectively), we have performed an additional analysis that extends these parameters to the general settings; it suggests that the buyer has more incentive to offer a restrictive RFQ to provide the demand visibility for the customized products, where the above transactional costs are expected to be higher. On the other hand, if the buyer can easily access the emergency sources in the case of shortfall (i.e., low c_I) or salvage the excess units without incurring too much cost (i.e., high c_S), as is the case for a number of commodity products, the buyer prefers a non-restrictive RFQ over a restrictive one.

The model presented in this study can be extended in various directions. We assumed here that supply chain firms are risk-neutral optimizers. One possibility would be to introduce risk aversion into the players' objectives. This extension would enable us to see the impact of the contracting decision on the risk characteristics. Another extension is to consider the impact of credible information sharing on supply chains with both upstream and downstream competitive structures. Moreover, considering a multi-period setting would allow us to capture the learning ad reputation effects. Lastly, we believe that the analysis of the impact of credible forecast sharing on competitive supply chain settings presents fruitful research opportunities and hope that our model will fuel future research in this endeavor.

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Notes

[1]In practice, the basic logic of a standard RFQ process has been implemented in various ways such as posting a "buying request" in online B2B market places, e.g., Alibaba.com (for sample buying requests, see AliSourcePro 2013), or initiating a "reverse auction" by using one of the sourcing softwares and services, e.g., Ariba, Iasta, etc. (Williams 2010). We also refer the readers to Johnson and Whang (2002) for an extensive review of e-procurement in SCM.

[2]The specific standard developed for RFQs by the United Nations is called REQOTE (see REQOTE (2002) for the detailed definition of minimum and maximum order quantities). This standard has been implemented by many enterprise resource planning softwares including SAP and Oracle. Particularly, Enterprise Applications Suite developed by Oracle enables the users to specify both *minimum* and *maximum* order quantities in RFQs—see User's Guide for Oracle® Purchasing Release 12 (Oracle 2006).

[3]The other top two reasons for not sharing forecast information cited by respondents were internal process change (60% of respondents) and cost of implementation (50% of respondents).

[4]Throughout the article, we use masculine and feminine pronouns for the buyer and suppliers, respectively.
[5]A notable exception is Maskin and Tirole (1992), where the authors characterize the set of equilibria by using a strategy space that consists of all contracts. However, they consider a restrictive case in which there is only one agent.
[6]"Two-point" (all-or-nothing) distributions are commonly used in asymmetric information literature for analytical tractability. Refer to Yang et al. (2009) and references therein for examples.
[7]The first-order stochastic dominance implies that *h*-type is more likely to have a high demand state than *l*-type. For the rigorous definition and treatment of stochastic orders, see Shaked and Shanthikumar (1994).
[8]We have performed additional analyses on similar models where the uncertainty is defined on the suppliers' capacities (as opposed to their yields) and shown that our results are not qualitatively affected by this modeling extension.
[9]Technically speaking, the insourcing option is equivalent to a lost sales demand model, where the buyer collects revenue p from each unit sold, and unmet demand is assumed to be lost. We can transform the insourcing option to a lost sales demand model by letting the buyer's marginal revenue be equal to the marginal insourcing cost, i.e., $p = c_I$.
[10]Essentially, the last assumption implies that the buyer never finds it optimal to set $q_U + q_R = 0$ in the equilibrium. For details, please refer to the Appendix. We would like to note that this assumption can be relaxed without affecting the qualitative nature of our results. The extended analysis is available from the authors upon request.
[11]Relative underage (resp., overage) cost represents the differential cost incurred by the buyer when he ordered from supplier U as opposed to from supplier R if demand turns out to be more (resp., less) than supply.
[12]Throughout the article, we use greater/smaller, higher/lower, and increasing/decreasing in the weak sense.
[13]The discussion for the other case C_U can be obtained by exchanging *l* type with *h* type.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1. Proofs of Propositions.